

Fall 1995

Problem 1 (Fa95). Let G be a group generated by n elements. Find an upper bound $N(n, k)$ for the number of subgroups H of G with the index $[G : H] = k$.

Problem 2 (Fa95). Let A be a finite subset of the unit disc in the plane, and let $N(A, r)$ be the set of points at distance $\leq r$ from A , where $0 < r < 1$. Show that the length of the boundary $N(A, r)$ is, at most, C/r for some constant C independent of A .

Problem 3 (Fa95). Find the radius of convergence R of the Taylor series about $z = 1$ of the function $f(z) = 1/(1 + z^2 + z^4 + z^6 + z^8 + z^{10})$. Express the answer in terms of real numbers and square roots only.

Problem 4 (Fa95). Let $F \subset E$ be a field and an extension. If two matrices in F are similar over E then they are similar over F .

Problem 5 (Fa95). Prove that $\mathbb{Q}[x, y]/\langle x^2 + y^2 - 1 \rangle$ is an integral domain and that its field of fractions is isomorphic to the field of rational functions $\mathbb{Q}(t)$.

Problem 6 (Fa95). Determine all real numbers $L > 1$ so that the boundary value problem

$$x^2 y''(x) + y(x) = 0 \quad 1 \leq x \leq L$$

$$y(1) = y(L) = 0$$

has a nonzero solution.

Problem 7 (Fa95). Let $f(z) = \sum_{n=0}^{\infty} f_n z^n$ and $g(z) = \sum_{n=0}^{\infty} g_n z^n$ be entire functions. Find a formula for the coefficients h_n in the Taylor expansion about $z = 0$ of

$$h(z) = \frac{1}{2\pi i} \int_{|w|=1} \frac{1}{w} f(w) g\left(\frac{z}{w}\right) dw$$

Problem 8 (Fa95, Fa18, Fa24). Show that an $n \times n$ matrix of complex numbers A satisfying

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|$$

for $1 \leq i \leq n$ must be invertible.

Problem 9 (Fa95). Let x_1 be a real number, $0 < x_1 < 1$, and define a sequence by $x_{n+1} = x_n - x_n^{n+1}$. Show that $\liminf_{n \rightarrow \infty} x_n > 0$.

Problem 10 (Sp87, Fa95). Let $\mathbb{F}$ be a field. Prove that every finite subgroup of the multiplicative group of nonzero elements of $\mathbb{F}$ is cyclic.

Problem 11 (Fa95). Let $f(z) = u(z) + iv(z)$ be holomorphic on $|z| < 1$, u and v real. Show that

$$\int_0^{2\pi} u(re^{i\theta})^2 d\theta = \int_0^{2\pi} v(re^{i\theta})^2 d\theta$$

for $0 < r < 1$ if $u(0)^2 = v(0)^2$.

Problem 12 (Fa95). Let f and f' be continuous on $[0, \infty)$ and $f(x) = 0$ for $x \geq 10^{10}$. Show that

$$\int_0^\infty f(x)^2 dx \leq 2 \sqrt{\int_0^\infty x^2 f(x)^2 dx} \sqrt{\int_0^\infty f'(x)^2 dx}$$

Problem 13 (Fa95). Show that

$$(1 + z + z^2 + \cdots + z^9) \cdot (1 + z^{10} + z^{20} + \cdots + z^{90}) \cdot (1 + z^{100} + z^{200} + \cdots + z^{900}) \cdots = \frac{1}{1-z}$$

for $|z| < 1$.

Problem 14 (Fa95). Let $f(x) \in \mathbb{Q}[x]$ be a polynomial with rational coefficients. Show that there is a $g(x) \in \mathbb{Q}[x]$, $g \neq 0$, such that $f(x)g(x) = a_2x^2 + a_3x^3 + a_5x^5 + \cdots + a_px^p$ is a polynomial in which only prime exponents appear.

Problem 15 (Fa95). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a C^∞ function. Assume that $f(x)$ has a local minimum at $x = 0$. Prove there is a disc centered on the y axis which lies above the graph of f and touches the graph at $(0, f(0))$.

Problem 16 (Fa95). Let A and B be nonsimilar $n \times n$ complex matrices with the same minimal and the same characteristic polynomial. Show that $n \geq 4$ and the minimal polynomial is not equal to the characteristic polynomial.

Problem 17 (Fa95). Let $f_1, f_2, \dots, f_n$ be continuous real valued functions on $[a, b]$. Show that the set $\{f_1, \dots, f_n\}$ is linearly dependent on $[a, b]$ if and only if

$$\det \left(\int_a^b f_i(x) f_j(x) \, dx \right) = 0$$

Problem 18 (Fa95). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a nonzero C^∞ function such that $f(x)f(y) = f\left(\sqrt{x^2 + y^2}\right)$ for all x and y and that $f(x) \rightarrow 0$ as $|x| \rightarrow \infty$.

1. Prove that f is an even function and that $f(0)$ is 1.
2. Prove that f satisfies the differential equation $f'(x) = f''(0)xf(x)$, and find the most general function satisfying the given conditions.